Beyond Financial Taxonomy: Structural Patterns in a Bluesky Ticker Co-Mention Network

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Abstract

With the growing popularity of online social networks, social media platforms have become important arenas for financial discussion. Viral trends, influential users, and market narratives may shape collective attention toward assets that are not closely related according to conventional financial classifications. This paper studies a large-scale ticker co-mention network constructed from approximately 25,000 multi-ticker Bluesky posts identified within a corpus of approximately 2.5 million retrieved posts, in which nodes represent financial tickers and edges connect tickers jointly mentioned in the same post. The resulting graph contains 6,243 nodes and 462,904 edges and exhibits a short average path length ($L \approx 2.50$), a high average clustering coefficient ($C \approx 0.482$, roughly 20× the Erdős–Rényi baseline), and mild positive degree assortativity ($r \approx 0.052$). We investigate the extent to which the observed co-mention topology aligns with official financial sector classifications. Community detection using Louvain, Label Propagation, and Greedy Modularity reveals weak agreement between detected communities and Yahoo Finance sector labels, including across different Louvain resolution parameters. We then evaluate held-out edge reconstruction using topological features such as Common Neighbors, Jaccard, Adamic-Adar, and Preferential Attachment, and a binary same-sector feature. Topological features account for nearly all reconstruction performance, while adding sector information improves AUROC by at most approximately 0.0007. Together, these results indicate that official sector taxonomy provides only limited additional information about the observed co-mention structure. The extensive cross-sector organization of the network is consistent with broader attention and narrative based discussion patterns, although such narratives are not measured directly in this study.

Keywords

Social Network Analysis, Bluesky, financial networks, ticker co-mentions, community detection, link prediction

1 Introduction

Over the past two decades, online social media platforms have become increasingly important channels for the production, diffusion, and discussion of financial information. Investors and other market participants can rapidly exchange opinions, interpretations, and reactions to market events, allowing collective attention to form around particular assets or themes. This increased accessibility has broadened participation in financial discussions, but it has also raised concerns about the influence of non-professional financial content creators, often referred to as "finfluencers" [5].

Project repositories: https://github.com/sna-unipi/sna-final-project-2026-2026_buccolini_fuschi_ja/tree/main

Conventional financial taxonomies organize companies according to characteristics such as economic sector and industry. Social-media discussions, however, need not follow the same boundaries. Assets from different sectors may be discussed together because they are simultaneously exposed to the same market event, macroeconomic development, technological theme, or broader pattern of online attention. A ticker co-mention network therefore provides a useful representation of how financial assets are associated within online discussions, independently of whether those associations correspond to established financial classifications.

Bluesky provides an appropriate setting for studying these patterns because its publicly accessible infrastructure allows the systematic collection of posts and the construction of reproducible social-network datasets. Rather than modeling relationships between users, this study focuses on relationships between financial tickers appearing in Bluesky posts.

We construct a ticker co-mention network in which nodes represent financial tickers and an undirected edge connects two tickers when they are mentioned together in at least one post. The resulting topology captures patterns of joint attention among financial instruments, rather than direct economic relationships between the corresponding assets.

The central research question of this work is to what extent the topology of the Bluesky co-mention network aligns with established financial sector classifications, and how much additional information sector membership provides beyond the network's local structure. Weak alignment with sector labels would indicate that the organization of online co-mentions cannot be adequately described by conventional sector taxonomy alone. Such a result would be consistent with the presence of broader cross-sector discussion themes, attention dynamics, and market narratives, although the semantic content of these narratives is not directly modeled in this study.

To investigate this question, we combine network characterization, community detection, and link-prediction methods. Community detection is used to evaluate whether communities identified from the network topology correspond to official financial sector labels. Link prediction is used in a static edge-reconstruction setting: observed edges are held out and reconstructed from topological features, allowing us to assess whether adding a same-sector indicator provides predictive information beyond local network structure. The task therefore evaluates reconstruction of hidden co-mention relationships within the observed network and should not be interpreted as forecasting future co-mentions.

Finally, we investigate whether highly central nodes connect heterogeneous financial sectors. We refer to these nodes as "narrative bridges" as an interpretive description of their cross-sector brokerage role. The analysis itself is based on structural centrality

and sector diversity, rather than on direct semantic identification of specific narratives.

By comparing detected communities with financial metadata, evaluating the incremental contribution of sector information in supervised edge reconstruction, and examining the cross-sector role of structural hubs, we quantify the extent to which official financial taxonomy is reflected in Bluesky co-mention patterns.

The remainder of the paper is organized as follows. Section 2 describes the data-collection and metadata-enrichment pipeline. Section 3 characterizes the main structural properties of the network. Sections 4 and 5 present the community-detection and link-prediction analyses, respectively. Section 6 investigates the role of highly central tickers as potential cross-sector structural bridges. The final section discusses the results, their limitations, and possible directions for future work.

2 Data Collection

The objective of the data-collection phase was to construct a large-scale network describing which financial tickers are jointly mentioned in Bluesky posts. Rather than modeling interactions between users, we focus on relationships between financial instruments and represent joint online attention through ticker co-occurrences.

The complete pipeline consists of three stages: (i) retrieving Bluesky posts through finance-related and ticker-seeded search queries, (ii) identifying valid financial cashtags and constructing the ticker co-mention network, and (iii) enriching the resulting ticker nodes with external financial metadata.

2.1 Selected Data Sources

The primary data source is the Bluesky social network, accessed through its public API [2]. A reference universe of tradable symbols was constructed from the Nasdaq Trader Symbol Directory [3], using both the `nasdaqlisted` and `otherlisted` files. Additional company-level metadata were subsequently retrieved from Yahoo Finance through the `yfinance` Python library [1].

2.1.1 Crawling Methodology and Assumptions. The Bluesky retrieval procedure was executed from July 7 at approximately 14:15 until July 8 at approximately 05:30. This interval refers to the execution of the data-collection process rather than to the publication dates of the retrieved posts: the search API was not restricted to posts created during this interval and could return earlier content.

The collection followed a query-based strategy. An initial set of 61 finance-related queries included general market terms, sector-related expressions, and frequently discussed financial cashtags. This stage produced 24,766 unique posts. To increase coverage of the financial-symbol universe, a second extraction used 10,000 ticker-seeded queries of the form `$TICKER`. For each query, up to five result pages were requested, with at most 100 posts per page. The crawler incorporated delays between requests, periodic checkpoints, and retry/backoff logic for temporary API failures. This second stage retrieved 2,487,835 unique posts. After merging the two collections and removing duplicates on the basis of the Bluesky post URI, the final retrieval corpus contained 2,498,033 unique posts.

Because the dataset was obtained through search queries rather than through an exhaustive stream of all Bluesky activity, it should be interpreted as a query-based sample of financial discussion on the platform rather than as a representative sample of all Bluesky posts.

2.1.2 Ticker Identification. Ticker recognition was restricted to explicit cashtag expressions, such as `$AAPL` or `$NVDA`, rather than arbitrary uppercase words. Candidate cashtags were extracted from post text using a regular expression and converted to uppercase before validation against the Nasdaq Trader reference universe.

The initial Nasdaq Trader files contained 12,931 unique listed symbols. After removing text issues and retaining symbol formats compatible with cashtag extraction, the broad whitelist contained 12,511 valid financial symbols. This broad universe intentionally includes not only common corporate equities, but also ETFs and other listed financial instruments. Consequently, nodes in the network are interpreted as financial tickers or listed instruments rather than exclusively as publicly traded companies.

For symbols containing dots or dashes, equivalent textual variants were mapped to a common canonical ticker when possible. Validation against the external whitelist substantially reduces false positives arising from arbitrary cashtag strings, although ticker-symbol ambiguity cannot be eliminated completely because the intended meaning of a cashtag is not inferred from the semantic content of the post.

After cashtag extraction, 116,461 unique posts contained at least one valid financial ticker, while 25,229 contained at least two valid tickers and could therefore generate at least one co-mention edge. Across all retrieved posts, 7,642 distinct valid tickers were observed. However, only tickers appearing in at least one multi-ticker post can participate in the co-mention graph, resulting in 6,243 network nodes.

2.1.3 Network Construction. For each post containing at least two valid tickers, duplicate ticker mentions within the same post were removed and all unordered pairs of distinct tickers were generated. Thus, a post containing m distinct tickers can generate up to

$$\binom{m}{2}$$

ticker pairs.

Repeated appearances of the same pair across different posts were counted during construction, producing an intermediate weighted representation in which the weight of an edge corresponds to the number of posts in which the two tickers were jointly mentioned. The analyses presented in this paper, however, use the corresponding undirected, unweighted simple graph: each distinct ticker pair is represented by a single edge regardless of the number of repeated co-mentions. The final unweighted network contains 6,243 nodes and 462,904 distinct edges.

This one-mode projection of multi-ticker posts has an important structural consequence: a post containing three or more tickers can mechanically create triangles and other dense local structures in the projected network. Therefore, part of the observed clustering may arise from the co-occurrence projection itself rather than exclusively from an independent tendency toward triadic closure. This issue is taken into account when interpreting the network-characterization results.

2.1.4 Financial Metadata. The network was enriched with external financial metadata retrieved programmatically through `yfinance` [1]. Since these data are retrieved from an external source at a specific point in time, rerunning the collection procedure at a later date may yield different metadata. The collected attributes include:

- ticker symbol;
- instrument or company name;
- market exchange;
- economic sector;
- industry.

Sector information was available for 5,091 of the 6,243 network nodes and unavailable for 1,152 nodes. Missing sector labels are particularly relevant for instruments that do not correspond to conventional corporate equities, including broad-market funds and other financial products for which the corporate sector taxonomy is not directly applicable. We therefore treat missing sector membership explicitly rather than assigning such instruments to an artificial sector.

The external metadata provide a financial taxonomy that is independent of the co-mention topology. Sector information is used in two complementary ways. First, community-detection results are evaluated by comparing topology-derived communities with official sector labels. Second, a binary same-sector indicator is introduced in the supervised edge-reconstruction analysis to measure whether sector membership provides additional predictive information beyond local topological features. Low sector alignment or limited incremental predictive value should therefore be interpreted as evidence that conventional sector taxonomy alone does not fully account for the observed co-mention organization, rather than as direct evidence of any specific narrative mechanism.

3 Network Characterization

After collapsing repeated co-mentions into unique undirected edges, the network contains 6,243 nodes and 462,904 edges, corresponding to an average degree $\langle k \rangle \approx 148.30$ and a density of approximately 0.0238. The graph contains no self-loops, since only co-mentions between distinct tickers are represented.

3.1 Connected Components

The network is composed of 16 connected components but is overwhelmingly dominated by a single giant connected component (GCC). The GCC contains 6,211 nodes, corresponding to 99.49% of all nodes, and 462,886 of the 462,904 edges (99.996%). The remaining 32 nodes are distributed across 15 small components: two components containing three tickers each and thirteen isolated pairs.

The concentration of almost all nodes and edges in the GCC indicates that the co-mention network is highly interconnected despite the presence of a small number of peripheral ticker groups. Broadly discussed instruments may contribute to this connectivity by linking otherwise distant regions of the network, although the specific discussion contexts responsible for these connections are not directly identified from topology alone.

Distance-based quantities, including shortest-path statistics and centrality measures requiring connectedness, are computed on the GCC. Analyses that do not require a connected graph use the full network unless otherwise specified. This distinction avoids undefined distances between disconnected nodes while preserving the complete topology whenever connectedness is not required.

3.2 Degree Distribution

Within the GCC, the degree distribution is strongly right-skewed (skewness ≈ 2.43), with a minimum degree of 1, a maximum of 1,846, a mean degree of approximately 149.05, and a median of only 64. The difference between the mean and median, together with the positive skewness, indicates substantial degree heterogeneity: most tickers have relatively limited connectivity, while a small number of nodes function as highly connected hubs.

The highest-degree nodes include major US equities and broad-market exchange-traded funds: NVDA (NVIDIA) (1,846), SPY (SPDR S&P 500 ETF) (1,809), TSLA (Tesla) (1,619), AAPL (Apple) (1,447), QQQ (Invesco QQQ Trust) (1,425), and AMD (Advanced Micro Devices) (1,389). Their high degrees indicate that, within the collected Bluesky sample, these tickers are co-mentioned with a particularly wide range of other financial instruments.

A power-law fit to the degree distribution yields an estimated exponent $\alpha \approx 1.98$ with $x_{\min} = 89$. However, a log-likelihood ratio comparison between a power-law and a log-normal model produces $R = -283.7$ with $p < 0.0001$. The negative likelihood ratio provides strong evidence in favor of the log-normal model relative to the power-law alternative. Therefore, although the upper tail appears approximately linear in a log-log representation, the relative statistical comparison does not support interpreting this visual pattern as evidence of a pure power-law degree distribution.

The network should consequently be described as highly heterogeneous and heavy-tailed rather than classified as scale-free on the basis of its degree distribution alone. The presence of extremely well-connected hubs does not by itself establish a preferential-attachment or scale-free generating mechanism. Instead, the observed degree heterogeneity is consistent with strongly uneven levels of visibility and joint attention across financial tickers.

Figure 1 illustrates this structure, with most nodes concentrated at relatively low degrees and a much smaller set of highly connected hubs.

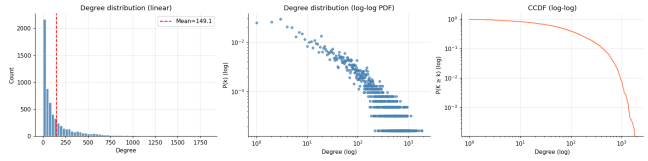


Figure 1: Degree distribution of the giant connected component, shown through the empirical histogram, log-log probability density, and complementary cumulative distribution function (CCDF).

3.3 Clustering and Local Structure

The average local clustering coefficient of the GCC is $C \approx 0.482$, while global transitivity is approximately 0.284. Both values are

substantially higher than the density-based Erdős–Rényi expectation,

$$C_{\mathrm{ER}} = p \approx \frac{\langle k \rangle}{N-1} \approx 0.024.$$

In particular, the average local clustering coefficient is approximately

$$\frac{C_{\mathrm{obs}}}{C_{\mathrm{ER}}} \approx 20$$

times the corresponding ER baseline. This indicates substantially stronger local triangle density than would be expected in an independent-edge random graph with comparable size and density.

This comparison must nevertheless be interpreted in light of the network-construction procedure. The ticker graph is a one-mode projection of multi-ticker posts: whenever a single post contains three or more distinct tickers, the projection can mechanically generate triangles among those tickers. Consequently, part of the observed clustering is induced by the co-occurrence construction itself and should not be interpreted entirely as independent triadic closure or thematic organization.

Clustering also tends to decrease with degree. Highly connected nodes such as SPY and QQQ have local clustering coefficients of approximately 0.09–0.15, whereas some lower-degree nodes belong to nearly complete local neighborhoods and can reach clustering coefficients close to 1. This pattern is compatible with high-degree hubs participating in multiple distinct regions of the network, while lower-degree nodes may be concentrated within more locally cohesive co-mention groups.

3.4 Centrality Analysis

Five centrality measures were computed on the GCC: degree centrality, closeness centrality, approximate betweenness centrality using $k = 500$ sampled pivot nodes, PageRank with damping parameter $\alpha = 0.85$, and eigenvector centrality.

NVDA, SPY, and TSLA rank among the most central nodes across several of these measures, confirming that they occupy prominent structural positions in the observed network. An important distinction emerges between degree and brokerage-related measures. NVDA has the highest degree, whereas SPY ranks first in betweenness centrality (approximately 0.053) and PageRank (approximately 0.0029). The high betweenness of SPY indicates that it lies on a comparatively large share of shortest paths and therefore connects multiple regions of the co-mention network. This structural result is consistent with a brokerage role, although the specific semantic contexts connected by SPY cannot be inferred from topology alone.

Broad-market instruments such as SPY, QQQ, and DIA also display substantial betweenness, suggesting that they contribute to connectivity between different ticker neighborhoods. Their role is examined more explicitly in Section 6, where structural centrality is combined with sector diversity.

Eigenvector centrality highlights a group of highly connected technology- and AI-adjacent assets, including NVDA, INTC, AMD, MU, and TSLA. High eigenvector centrality indicates that these nodes are not only well connected, but are also connected to other structurally important nodes. At the same time, the comparatively low local clustering coefficients of several major hubs suggest that their connectivity extends across multiple neighborhoods rather than being restricted to a single tightly closed group.

3.5 Degree Assortativity

The degree assortativity coefficient is $r \approx +0.052$, indicating weak positive degree assortativity at the global level. Thus, high-degree tickers show a slight tendency to connect to other high-degree tickers more frequently than would be expected under neutral degree mixing.

The relationship is more nuanced when examined through the average-neighbor-degree function $k_{\mathrm{nn}}(k)$ in Figure 2. Average neighbor degree generally increases across much of the degree range, but the relationship is not strictly monotonic and weakens among the most highly connected hubs. This suggests the coexistence of a relatively interconnected high-degree core and a set of major hubs that also connect extensively to lower-degree peripheral nodes.

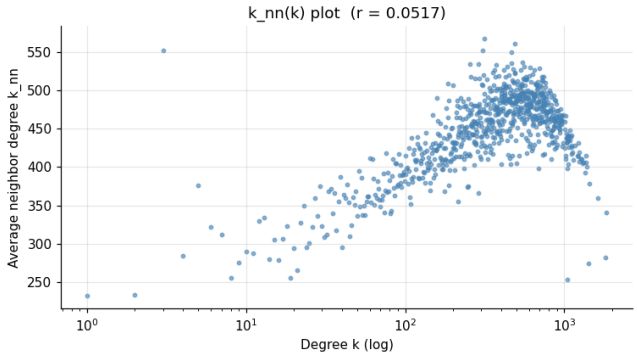


Figure 2: Average neighbor degree $k_{\mathrm{nn}}(k)$ as a function of node degree. The overall pattern is consistent with weak positive assortativity, with a non-monotonic behavior among the highest-degree nodes.

3.6 Comparison with ER and BA Synthetic Models

To provide descriptive reference points for the empirical network, we generated two synthetic graphs with the same number of nodes ($N = 6{,}243$) and approximately comparable numbers of edges. The first is an Erdős–Rényi graph $G(N, p)$, with p selected to reproduce approximately the empirical density, yielding $E = 461{,}363$. The second is a Barabási–Albert preferential-attachment graph with $m = 74$ edges added for each new node, yielding $E = 456{,}506$. The comparison is summarized in Table 1.

The empirical network differs from both synthetic references in several respects. First, its average clustering coefficient (0.482) is approximately 20 times that of the ER graph (0.024) and about 7.5 times that of the BA graph (0.064). Neither synthetic model reproduces the high level of local closure observed in the empirical projection. However, as discussed above, this difference cannot be attributed exclusively to social or thematic mechanisms because the projection of multi-ticker posts itself contributes mechanically to triangle formation.

Second, the average shortest-path length of the empirical GCC ($L \approx 2.50$) is moderately larger than those of the ER (2.01) and BA (2.03) reference graphs. The empirical network therefore remains

Table 1: Summary statistics of the Bluesky ticker co-mention network and the generated Erdős–Rényi (ER) and Barabási–Albert (BA) reference graphs.

Metric	Real (Bluesky)	ER	BA
N	6,243	6,243	6,243
E	462,904	461,363	456,506
$\langle k \rangle$	148.30	147.80	146.25
Density	0.02376	0.02368	0.02343
Avg clustering C	0.482	0.024	0.064
Transitivity	0.284	0.024	0.065
Assortativity r	+0.052	−0.001	+0.004
# Components	16	1	1
GCC fraction	99.49%	100%	100%
Avg path L	2.50	2.01	2.03

highly compact, although it is less globally efficient than these synthetic graphs of comparable size and density.

Third, the empirical graph exhibits mild positive degree assortativity ($r \approx +0.052$), with a value broadly comparable to those of the generated ER and BA graphs, which are close to zero.

Taken together, the network displays a combination of short path lengths and clustering substantially above the ER reference, a pattern consistent with small-world-like organization. Relative to the ER benchmark,

$$\frac{C_{\text{obs}}}{C_{\text{ER}}} \approx 20, \qquad \frac{L_{\text{obs}}}{L_{\text{ER}}} \approx 1.24,$$

showing that the empirical network combines substantially elevated clustering with only a moderate increase in path length.

The ER and BA graphs should nevertheless be interpreted as descriptive reference models rather than definitive null models. In particular, they do not preserve the empirical degree sequence or the bipartite post–ticker structure from which the co-mention network is projected. The observed structural differences therefore motivate further analysis of community organization and link reconstruction, but do not by themselves identify the mechanisms responsible for cross-sector co-mention patterns.

4 Task 1: Community Detection

The first analytical task investigates the modular organization of the Bluesky ticker co-mention network. The guiding question is not merely which communities the algorithms can find, but also what the communities represent. Particularly, it has been analyzed the existence of an alignment with the financial taxonomy encoded in the Yahoo finance sector labels or of narrative forces that govern social media discussions.

4.1 Preliminary checks

The algorithms were applied to the known-sector subgraph, comprising 5,091 nodes. Initial analyses suggest that financial sectors exhibit weak signal characteristics prior to community detection, confirming that the subgraph is not artificially sector-driven. Three diagnostics establish this:

- **Sector assortativity.** The attribute assortativity coefficient on the known-sector subgraph is $r_{\text{sector}} = 0.058$, barely above zero. While positive, this value is far below what one would expect from a graph genuinely organized along sector lines. The mild positive value confirms that sector homophily exists but is a weak force compared with the cross-sector dynamics that dominate the co-mention topology.
- **Modularity of the official partition.** Treating the 10 Yahoo Finance sectors[0] as a fixed partition and computing its modularity on the known-sector subgraph yields $Q_{\text{sectors}} = 0.051$. A modularity this low indicates that the sector boundaries induce almost no more internal density than a random partition of the same sizes would.
- **Same-sector edge rate.** Only 18.3% of known-known edges connect tickers within the same sector, against a random baseline of 12.8% implied by the sector size distribution. The signal is statistically real but economically small: sector membership does not significantly raise the probability of a co-mention edge.

The network was formed by co-occurrence in posts, not by financial proximity, so the topology is primarily shaped by the attention dynamics of Bluesky users rather than by the economic structure of the companies they discuss.

4.2 Community Detection Algorithms

Three classical community detection algorithms have been applied using the CDlib library [4]:

- **Louvain**
- **Label Propagation**
- **Greedy Modularity**

To evaluate the detected community structures, three main metrics were considered: modularity, Sector Normalized Mutual Information (Sector NMI), and weighted sector purity.

Modularity was computed using CDlib's `newman_girvan_modularity` function to assess the internal density of the detected partitions relative to a random null model.

Sector NMI measures the information-theoretic alignment between the algorithmically detected community structure and the ground-truth Yahoo Finance sector partition. It quantifies the mutual information shared between the two partitions.

Weighted sector purity measures the extent to which each detected community is dominated by a single sector, scaled by community size:

$$P_w = \frac{1}{N} \sum_c |c| \cdot \frac{\max_s |\{v \in c : \text{sector}(v) = s\}|}{|c|}, \tag{1}$$

where the sum runs over all detected communities c, $|c|$ is the size of community c, and N is the total number of labeled nodes. Weighted purity is preferred over a simple average because small or trivial communities naturally exhibit high purity by size alone, which would artificially inflate an unweighted mean.

[0]The 10 sectors are: Healthcare, Technology, Industrials, Fanancial Services, Consumer Cyclical, Basic Materials, Communication Services, Real Estate, Consumer Defensive, and Energy.

4.3 Results

Table 2 summarises the output of all three algorithms.

Table 2: Community detection results on the known-sector subgraph.

Algorithm	Comm.	Mod.	Sec. NMI	Weigh. Pur.
Louvain	70	0.212	0.049	0.274
Label Prop.	62	0.000	0.024	0.214
Greedy Mod.	79	0.198	0.048	0.270

The results reveal a clear split between topological quality and semantic alignment. Louvain and Greedy Modularity both find partitions with modularity around 0.20-0.21: these are non-trivial, well-separated communities in the graph-structural sense.

The two modularity-based algorithms find a range of 70-79 communities, a number far larger than the 10 sectors, already hinting that the network is partitioned along finer, non-sector lines. Despite their solid modularity, Louvain and Greedy Modularity achieve sector NMI below 0.05 and weighted purity of only ≈0.27. Both values are very close to the performance of Label Propagation (NMI = 0.023, purity = 0.214), an algorithm that effectively fails to find any structure at all. The implication is stark: a sophisticated modularity-maximizing algorithm and a near-random label diffuser produce sector-alignment scores in the same ballpark. This shows that sector alignment is not an artifact of any one method; rather, it reflects an intrinsic property of the data.

Table 3: Comparison of community detection algorithms against official sector labels.

Algorithm	Sector NMI	Sector AMI
Louvain	0.048767	0.030631
Label Propagation	0.023822	-0.000139
Greedy Modularity	0.048464	0.023442

The evaluation metrics presented in Table 3 highlight that Louvain and Greedy Modularity obtain similarly low alignment with sector labels, both yielding comparable Sector NMI (≈ 0.048) and AMI values. This high level of agreement confirms that the identified community partition represents a stable and reproducible topological feature of the network, rather than an artifact of a specific algorithm.

Conversely, Label Propagation diverges substantially, showing a near-zero (and slightly negative) Sector AMI. This result indicates that its detected partition is virtually indistinguishable from a random assignment relative to the official sectors, making modularity-maximization methods significantly more suitable for this graph structure.

4.4 Resolution Sweep Analysis

To test whether poor sector alignment stems from miscalibrated resolution, we varied Louvain's resolution parameter $\gamma \in [0.5, 12]$. Increasing γ improves sector alignment, raising Sector NMI from 0.050 ($\gamma = 1.0$, 70 communities) to 0.191 ($\gamma = 12.0$, 860 communities) and weighted purity from 0.277 to 0.429. However, modularity collapses from 0.213 to 0.059.

This reveals a fundamental trade-off: high resolution artificially inflates sector metrics via over-segmentation into micro-clusters (e.g., small groups of manufacturing or energy stocks), destroying the coherent large-scale network structure. Thus, low sector alignment is not a calibration artifact; recovering Yahoo sectors requires either heavy over-partitioning or accepting that sectors do not dictate the network topology.

4.5 Conceptual Interpretation

The joint evidence confirms that the Bluesky co-mention network is not financially organized. Topological community algorithms fail to recover GICS sectors because co-mentions are driven by thematic narratives rather than fundamental industry similarity. Users frequently pair tickers across sectors within shared discussion contexts—such as macro interest-rate outlooks (SPY, TLT, bank/real estate stocks) or AI narratives (NVDA, MSFT, PLTR).

Consequently, community detection identifies 'discussion clusters' rather than 'sector clusters'. The 70 communities that Louvain identifies likely correspond to coherent discussion topics (macro/rates, AI-related tech, crypto-adjacent equities, energy macro, speculative small-caps, etc.), topics that cut across the official financial taxonomy because social media narratives are thematic, not sector-aware. This number is the quantitative core of our answer to the research question: topology on Bluesky is not a proxy for financial taxonomy. Figure 3 illustrates this cross-community connectivity through the ego-network of NVDA. The central ticker is connected to nodes belonging to multiple topology-derived communities, highlighting the broad structural reach of highly connected assets within the co-mention network.

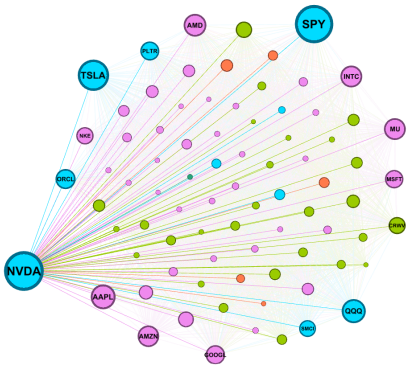


Figure 3: Ego-network centered on NVDA. Node colors represent topology-derived communities, while edges represent observed co-mentions within the displayed ego-network. The visualization illustrates the connections of a high-degree ticker across multiple detected communities.

5 Task 2: Supervised Link Prediction with Sector Information

The second analytical task investigates how effectively observed co-mention edges can be reconstructed from network topology and

whether financial-sector information provides additional predictive information.

Throughout this section, link prediction refers to held-out edge reconstruction within the same static co-mention graph and should not be interpreted as forecasting future co-mentions.

As a motivating baseline, the four classical local-topology scores introduced in class (Common Neighbors, Jaccard, Adamic-Adar, and Preferential Attachment) achieve AUROC between 0.91 and 0.97 under uniform negative sampling on this graph, confirming that local neighborhood overlap is a strong reconstruction signal. The supervised experiment extends this baseline by asking whether financial-sector membership provides additional discriminative information beyond what topology already captures.

5.1 Algorithms application and modelling

Because the same-sector feature is only defined for nodes with available sector metadata, candidate labeled pairs were restricted to the 5,091 nodes with known sector information. These nodes participate in 431,862 observed known-known edges.

The topology-only and topology-plus-sector models are trained and evaluated on exactly the same candidate pairs, ensuring that differences between the two models cannot be attributed to missing sector metadata.

A nested edge holdout was used to prevent direct leakage of labeled positive edges into the feature graph. First, 20% of the 431,862 known-known edges, corresponding to 86,372 edges, were reserved as final test positives. From the remaining edges, a further 20%, corresponding to 69,098 edges or approximately 16% of the original known-known set, were reserved as training positives. The remaining 276,392 known-known edges formed part of the structural context.

In addition, the 31,042 observed edges having at least one unknown-sector endpoint were retained because they still provide valid topological information. The resulting feature graph therefore contains 307,434 edges. None of the training-positive or test-positive edges was included directly in the graph from which the topological features were computed.

This procedure prevents direct edge leakage, but it does not eliminate the projection-induced dependence described above: other edges originating from the same multi-ticker post may remain in the structural graph.

For both the training and test positive sets, an equal number of true non-edges was sampled from pairs that are not connected in the complete observed graph. Two negative-sampling strategies were considered:

(1) **Uniform negatives**: non-edges sampled randomly among pairs of known-sector nodes;
(2) **Degree-matched negatives**: nodes were assigned to degree bins and, for each positive edge, a non-edge was sampled whose endpoints belong to corresponding degree bins.

Degree matching reduces the extent to which the classifier can distinguish positive and negative pairs simply from endpoint popularity. Training and test negative sets were kept disjoint.

The supervised classifier is logistic regression with standardized numerical inputs. Three feature sets were evaluated:

- **Sector only**: a binary indicator equal to one when the two endpoints belong to the same financial sector;
- **Topology only**: Common Neighbors, Jaccard, Adamic-Adar, and Preferential Attachment;
- **Topology + sector**: the four topological features combined with the same-sector indicator.

Table 4: Performance of supervised logistic-regression models under uniform and degree-matched negative sampling.

Negative sampling	Model	AUROC	AUPRC
Uniform	Sector only	0.5283	0.5167
Uniform	Topology only	0.9744	0.9716
Uniform	Topology + sector	0.9747	0.9718
Degree-matched	Sector only	0.5322	0.5195
Degree-matched	Topology only	0.8738	0.8755
Degree-matched	Topology + sector	0.8745	0.8762

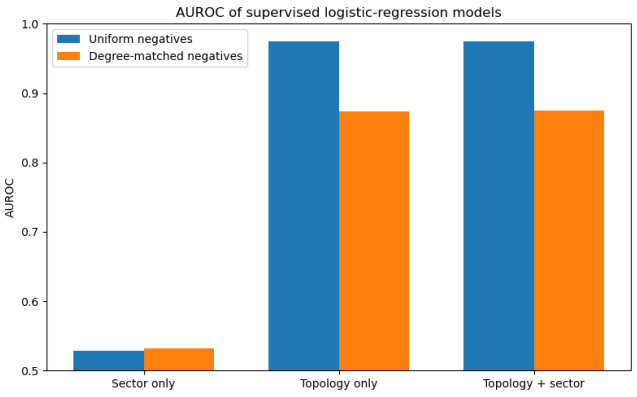


Figure 4: AUROC of supervised logistic-regression models using sector-only, topology-only, and combined feature sets under uniform and degree-matched negative sampling.

Table 4 and Figure 4 show that the same-sector indicator contains only weak discriminative information when used in isolation. The sector-only model reaches AUROC 0.5283 with uniform negatives and 0.5322 with degree-matched negatives, only slightly above the random baseline of 0.5.

This result is consistent with the limited difference in same-sector frequency between observed and non-edge pairs. Approximately 18.38% of positive test edges connect tickers belonging to the same sector, compared with 12.72% of uniformly sampled negative pairs and 11.93% of degree-matched negative pairs. Thus, the binary sector indicator provides some separation between the two classes, but most observed co-mention edges (approximately 81.62%) still connect tickers belonging to different sectors.

Topological features account for nearly all of the reconstruction performance. Under uniform negative sampling, adding the same-sector feature increases AUROC from 0.9744 to 0.9747 ($\Delta = 0.0003$) and AUPRC from 0.9716 to 0.9718. Under degree-matched sampling,

AUROC increases from 0.8738 to 0.8745 ($\Delta = 0.0007$), while AUPRC increases from 0.8755 to 0.8762.

The incremental improvement associated with sector information is therefore positive in the reported experiment but practically negligible relative to the contribution of network topology. Because the holdout and negative-sampling procedures are stochastic and confidence intervals across repeated random splits were not estimated, differences of this magnitude should not be interpreted as evidence of a statistically significant improvement.

The supervised results are consistent with the community-detection analysis. Sector membership is not entirely unrelated to co-mention structure, but it contributes little additional reconstruction information once local topology is known. Shared neighborhoods and normalized neighborhood overlap provide substantially stronger discrimination between observed edges and sampled non-edges than the binary same-sector feature.

This finding supports the conclusion that official sector taxonomy alone does not adequately describe the observed co-mention organization. Broader cross-sector attention patterns and market narratives are plausible explanations for the remaining structure, but they are not directly measured by the present reconstruction experiment.

Finally, both the supervised and unsupervised analyses remain static edge-reconstruction experiments. The balanced positive–negative evaluation sets are useful for comparing methods, but they do not reproduce the true prevalence of non-edges in the complete network. Moreover, because no temporal train–test evaluation is performed, the reported AUROC and AUPRC values quantify recovery of hidden edges in the observed graph rather than the ability to predict future Bluesky co-mentions.

6 Task 3: Open Question: Do Structural Hubs Act as Cross-Sector Bridges?

The previous analyses show that the Bluesky co-mention network contains substantial cross-sector structure that is only weakly aligned with official financial sector labels. This motivates a complementary question: do highly central tickers occupy structural positions that connect otherwise heterogeneous financial sectors?

We refer to such nodes as "narrative bridges" as an interpretive label for their potential role in connecting cross-sector discussions. The analysis itself, however, does not identify or classify the semantic content of specific narratives. Instead, it measures two observable structural properties: brokerage within the network and diversity in the sector composition of a node's neighbors.

6.1 Operationalizing Cross-Sector Bridges

A structural bridge is operationally defined as a ticker that combines high network centrality with a neighborhood spanning multiple financial sectors. We first identified structural hubs as nodes belonging to the top 5% of the degree distribution, corresponding to degree ≥ 606. This threshold yields 315 hub tickers. The 5% cutoff is an operational choice used to distinguish the most highly connected nodes rather than a theoretically unique definition of a hub.

For each node i with at least five neighbors having known sector information, we calculated two complementary quantities:

(1) **Approximate Betweenness Centrality (B_i):** computed on the giant connected component using $k = 500$ sampled pivot nodes. Betweenness measures the extent to which a node lies on shortest paths between other nodes and therefore provides an approximation of its structural brokerage role.

(2) **Sector Participation Coefficient (P_i):** defined as

$$P_i = 1 - \sum_s \left(\frac{k_{i,s}}{k_i} \right)^2 ,$$

where k_i is the number of neighbors of node i with known sector information and $k_{i,s}$ is the number of those neighbors belonging to sector s.

The participation coefficient approaches zero when a node's known-sector neighbors are concentrated within a single sector and increases as its neighborhood becomes more evenly distributed across multiple sectors.

For descriptive ranking purposes, we additionally defined a composite Bridge Score by summing the standardized values of betweenness centrality and sector participation:

$$\text{BridgeScore}_i = Z(B_i) + Z(P_i).$$

The composite score should be interpreted as a ranking device rather than as an independently calibrated measure of narrative influence. In particular, because betweenness centrality is strongly right-skewed, extremely central nodes can contribute disproportionately to the resulting score.

6.2 Sector Participation and Bridge Scores

Table 5 reports the ten highest-ranked tickers according to the composite Bridge Score. The ranking is dominated by broad-market instruments such as SPY, QQQ, and DIA, together with highly connected large-cap equities including NVDA, TSLA, AAPL, MSFT, AMD, and AMZN.

Among the individual equities for which a sector label is available, approximately 78%–87% of known-sector edges connect to tickers belonging to a different sector. This indicates that these highly central assets are embedded in strongly cross-sector neighborhoods rather than being connected predominantly within their own official sector.

Comparing hubs with the remaining nodes also reveals substantial structural differences. The 315 hubs have a mean sector participation coefficient of 0.8660, compared with 0.8020 among non-hubs. Their mean approximate betweenness centrality is also substantially larger (0.00217 compared with 0.00015). These results show that high-degree nodes tend to occupy more central positions and to connect to more sector-diverse neighborhoods.

However, this comparison should not be interpreted causally: high participation may partly arise mechanically because high-degree nodes have more opportunities to connect across multiple sectors. The permutation analysis in the following subsection is therefore used to evaluate whether the observed sector diversity differs from a randomized sector-label baseline.

Table 5: Top 10 tickers ranked by the composite Bridge Score. Cross-sector ratio represents the percentage of known-sector edges connecting an equity to a sector different from its own. The ratio is not defined for instruments without a conventional sector label.

Ticker	Sector	Degree	Betweenness	Cross-Sector	Bridge Score
SPY	*None (ETF)*	1,809	0.0527	*N/A*	47.49
QQQ	*None (ETF)*	1,425	0.0312	*N/A*	28.17
NVDA	Technology	1,846	0.0214	80.1%	19.51
TSLA	Consumer Cyclical	1,619	0.0173	87.2%	15.92
DIA	*None (ETF)*	1,046	0.0155	*N/A*	13.95
AAPL	Technology	1,447	0.0090	79.9%	8.53
MSFT	Technology	1,274	0.0078	78.1%	7.33
AMD	Technology	1,389	0.0072	78.8%	6.82
AMZN	Consumer Cyclical	1,357	0.0069	86.5%	6.57
BTC	*None*	706	0.0068	*N/A*	6.47

6.3 Sector-Label Permutation Analysis

To assess whether the high sector participation of structural hubs can be explained solely by their large and heterogeneous neighborhoods, we compared the empirical participation coefficients with a randomized sector-label baseline. Sector labels were randomly shuffled 100 times while keeping the network topology fixed, and the participation coefficients were recomputed after each permutation.

The randomized networks produced a mean hub participation coefficient of 0.8705, slightly higher than the empirical hub mean of 0.8660. The same pattern is more pronounced when considering all eligible nodes, for which the permuted mean participation is 0.8466 compared with an empirical mean of 0.8059.

These results provide an important qualification to the absolute cross-sector diversity observed among hubs. Although hubs have very high participation coefficients, they are not more sector-diverse than expected under the randomized sector-label baseline. Instead, their empirical participation is slightly lower than the null expectation. This pattern is consistent with the presence of residual sector homophily: sector membership retains some association with network connectivity even within an otherwise highly cross-sector graph.

Because only 100 permutations were performed and no empirical permutation p-value or confidence interval is reported, this comparison should be interpreted as a descriptive null-model analysis rather than as evidence of formal statistical significance.

Figure 5 summarizes the relationship between structural brokerage and sector diversity. The first panel compares betweenness centrality with participation coefficient, the second shows the distribution of the composite Bridge Score for hubs and non-hubs, and the third compares the empirical mean hub participation with its randomized sector-label distribution.

Overall, the analysis shows that the most highly connected tickers occupy structurally important positions and maintain highly sector-diverse neighborhoods. Broad-market instruments and highly visible equities therefore act as cross-sector structural bridges within the observed co-mention network. At the same time, the permutation comparison indicates that sector structure is not completely absent: empirical sector mixing is slightly more homogeneous than would be expected after randomizing sector labels.

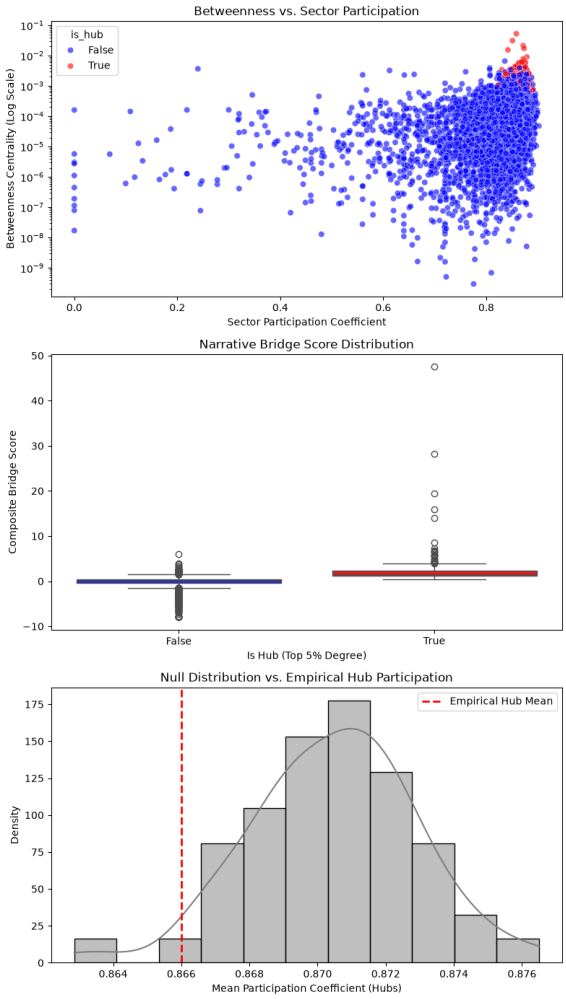


Figure 5: Cross-sector bridge analysis. (Top) Approximate betweenness centrality versus sector participation coefficient, with structural hubs distinguished from non-hubs. (Center) Distribution of the composite Bridge Score for hubs and non-hubs. (Bottom) Randomized sector-label distribution of mean hub participation compared with the empirical value.

These findings are consistent with a network in which highly central assets connect discussions spanning multiple financial sectors. However, the structural analysis alone cannot determine whether those connections are generated by specific market narratives, macroeconomic themes, speculative attention, or other mechanisms. Identifying such mechanisms would require combining the network with semantic analysis of the underlying Bluesky posts.

7 Discussion and Conclusion

In this paper, we investigated the structural organization of financial discussions on Bluesky through a large-scale ticker co-mention network. The main objective was to assess the extent to which the observed co-mention topology aligns with official financial sector

classifications and whether sector information provides explanatory value beyond local network structure. Rather than directly measuring social narratives or hype, the analysis evaluates whether conventional financial taxonomy is sufficient to account for the organization of ticker co-mentions.

The network characterization reveals a highly connected and heterogeneous graph dominated by a small number of high-degree hubs. The empirical network combines short average path lengths with clustering substantially higher than the Erdős–Rényi reference graph, a pattern consistent with small-world-like organization. However, the high clustering must be interpreted cautiously because the one-mode projection of multi-ticker posts can mechanically generate triangles. The degree distribution is strongly heterogeneous, but the statistical comparison between power-law and log-normal models does not support describing the network as purely scale-free.

The community-detection analysis provides the first indication that official sector taxonomy captures only a limited component of the observed network structure. On the known-sector subgraph, Louvain and Greedy Modularity identify non-trivial topological partitions, but their agreement with Yahoo Finance sector labels remains very weak, with NMI below 0.05 at the standard resolution. Increasing the Louvain resolution produces higher sector alignment only at the cost of strong fragmentation into hundreds of small communities. Sector membership therefore contributes some structural organization, but topology-derived communities do not closely reproduce the official financial sector partition.

The link-prediction analysis reaches a consistent conclusion from a different perspective. Local topological features, particularly Jaccard, Common Neighbors, and Adamic-Adar, provide strong discrimination between held-out observed edges and sampled non-edges. Their performance decreases substantially when more challenging negative-sampling strategies are introduced, showing that uniform negative sampling can produce an optimistic reconstruction task. In the supervised experiment, adding a binary same-sector feature increases AUROC from 0.9744 to 0.9747 under uniform sampling and from 0.8738 to 0.8745 under degree matching. These improvements are practically negligible relative to the contribution of network topology. Because confidence intervals across repeated random splits were not estimated, differences of this magnitude should not be interpreted as evidence of statistically significant improvement.

The open-question analysis further shows that the most highly connected tickers occupy important brokerage positions and maintain highly sector-diverse neighborhoods. Broad-market instruments such as SPY and QQQ, together with highly connected large-cap equities, rank among the strongest cross-sector structural bridges. The sector-label permutation analysis, however, provides an important qualification: the empirical mean participation coefficient of hubs (0.8660) is slightly lower than the randomized mean (0.8705). Thus, hubs are highly cross-sector in absolute terms, but sector organization is not completely absent. Instead, the results are consistent with weak residual sector homophily embedded within a network characterized by extensive cross-sector connectivity.

Taken together, the three analytical tasks provide a coherent picture. Official financial sector taxonomy is reflected in the network, but only weakly: sector assortativity is positive but small, topology-derived communities exhibit limited agreement with sector labels, the same-sector feature adds almost no reconstruction performance beyond topology, and highly central nodes connect broadly across sector boundaries. These findings indicate that conventional sector classification alone is insufficient to explain the observed organization of Bluesky ticker co-mentions.

The extensive cross-sector structure is consistent with financial discussions being organized around broader patterns of shared attention, market-wide events, and narratives that cut across traditional sector boundaries. However, these mechanisms are not directly identified by the present analysis. The study does not perform semantic topic modeling, sentiment analysis, or causal inference, and network topology alone cannot determine why two tickers are discussed together. The term "narrative bridge" should therefore be understood as an interpretive description of cross-sector structural brokerage rather than as direct evidence of a specific underlying narrative.

Several additional limitations should be considered. First, the dataset is a query-based sample of Bluesky financial discussion rather than an exhaustive representation of all platform activity. Second, the network is an unweighted one-mode projection of multi-ticker posts, which removes information about repeated co-mention frequency and can induce local clique structure. Third, sector labels are unavailable for a substantial subset of nodes, including several structurally important broad-market instruments. Fourth, the link-prediction experiments reconstruct randomly held-out edges from a static graph and therefore do not evaluate genuinely future co-mentions. Finally, random edge holdout can retain other edges generated by the same original multi-ticker post, potentially making local topological predictors optimistic.

These limitations also suggest several directions for future work. A temporal extension could construct training graphs from earlier posts and evaluate co-mentions appearing later, allowing genuine future link prediction and analysis of how network structure evolves around market events. Retaining co-mention frequencies would enable weighted-network analysis, while post-level or temporal train-test splits could reduce dependencies introduced by one-mode projection. Semantic analysis of the underlying posts, including topic modeling and sentiment analysis, could directly identify the narratives associated with cross-sector connections. Finally, combining the co-mention network with external financial variables such as returns, volatility, trading volume, or factor exposures could test whether online co-mention structure is associated with observable market relationships rather than inferring such effects from topology alone.

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